

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I Dharsan . B 192312150 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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CO3 ATI - Case Study

1) Cybersecurity Intrusion Detection System

a) Propositional logic representation:

The cybersecurity system uses propositional logic to represent security event and alerts.

Let:-

* F = Failed

* U = unusual IP

* O = odd logintime

* H = High data transfer

* D = possible data Exfiltration

Rules:

$$1. F \wedge U \rightarrow B$$

$$2. F \wedge O \rightarrow C$$

$$3. U \wedge H \rightarrow D$$

These rules allow the agent to infer possible security threats from network events

b) Forward chaining

Initial Facts: F, U, H

Step 1:- $F \wedge U \rightarrow B$

Both conditions are true

$\therefore B = \text{possible Brute Force Attack}$

Step 2:

$$U \wedge H \rightarrow D$$

Both conditions are true

$\therefore D = \text{possible Data Exfiltration}$

2) Airport priority landing system

a) unification

* Rule: $\text{Aircraft}(x) \wedge \text{EmergencyDeclared}(x) \rightarrow \text{priority clearance}(x)$.

* Facts: $\text{Aircraft}(\text{Flight } z), \text{EmergencyDeclared}(\text{Flight } z)$

Substitute $x = \text{Flight } z$.

$\therefore \text{priority clearance}(\text{Flight } z)$

b) Resolution

* Rule 2: $\text{priority clearance}(x) \rightarrow \text{Runway Reserved}(x)$.

$\therefore \text{Runway Reserved}(\text{Flight } z)$.

Rule 3 gives:-

* $\text{Aircraft}(y) \wedge \text{Queued After}(y, z) \wedge \text{RunwayReserved}(z) \rightarrow \text{Delay}(y)$

Therefore: $\text{Delay}(\text{Flight } y)$ is Proved.

c) Simultaneous Emergencies

* Current KB can identify multiple emergency aircraft.

* It does not specify how to share one runway.

* Additional rules needed!

- only one aircraft uses a runway at a time
- Higher emergency severity gets priority.
- Queue order resolves equal emergencies.

* conclusion: Additional runway-conflict rules are required.

3) Industrial Equipment Maintenance

a) CNF

- * P1: Machine overheating $\rightarrow$ Maintenance needed.
- * P2: $\neg$ maintenance needed $\vee \neg$ High priority line $\vee$ schedule Immediate Repair
- * P3: Schedule Immediate Repair $\vee \neg$ High priority line $\vee$ production Risk.
- * Facts: Machine overheating, High priority line

b) Resolution Refutation

- * Goal! Schedule Immediate Repair
- * Machine overheating + P1 $\rightarrow$ maintenance needed.
- * Maintenance needed + High priority line + P2 $\rightarrow$ schedule Immediate Repair
- * Schedule Immediate Repair + $\neg$ Schedule Immediate Repair $\rightarrow$ Empty clause.

c) Removing machine overheating

- * P1 cannot be activated
- * Maintenance needed cannot be derived.
- * Schedule Immediate Repair cannot be derived.
- * High priority line remains true.
- * $\neg$ schedule Immediate Repair is not given
- * Therefore, production Risk cannot be concluded.

4) Minefield Robot:

a) Belief State:

Initial percepts:

- vibration $\in [2, 1]$
- Smoke $\in [1, 1]$
- Beacon $\in [2, 2]$

Derived facts:

- Mine is adjacent to $[2, 1]$
- Fire is adjacent to $[1, 1]$
- Safe-zone marker is at $[2, 2]$

b) Forward chaining

* Exact mine / fire locations are not known from the given rules.

c) Path Safety

* Path: $[1, 1] \rightarrow [2, 1] \rightarrow [2, 2]$

* $[1, 1]$: smoke detected $\rightarrow$ fire danger nearby.

* $[2, 1]$: vibration detected $\rightarrow$ mine danger nearby

* $[2, 2]$: safe zone marker exists, but absence of mine/fire is not proven.

Conclusion:-

The complete path cannot be confirmed safe with the current knowledge base.